ETN Slovenia Real Estate Analyser

Repository-Specific Statistical and Machine Learning Guide

Audit-first implementation specification for Codex

Repository
matkopolo/NepremicninskiAgent

Branch: main
Inspected commit: 70b5ec82e964

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> **Purpose: tell Codex what already exists, what must be verified, and what should be corrected or added without duplicating the current architecture.**

Executive decision

> **This repository is not a greenfield analyser. Codex must begin with a repository audit and must not rebuild the analytical systems that are already present.**

The project already implements a substantial analytics platform: raw ETN ingestion, transaction-grain analytical facts, PostGIS geography, robust outliers, a layered anomaly queue, hedonic Ridge models, Isolation Forest scoring, prediction ranges, hedonic indices, repeat-transaction matching, a Go API and a Next.js dashboard.

The most valuable next work is methodological correction and extension: honest temporal/spatial validation, calibrated prediction intervals, accurate index naming and vintages, explicit buildable-land analysis, missingness/coverage reporting, near-duplicate detection, bootstrap uncertainty and segment-level rental yield.

Codex operating rule

1. Read the methodology and repository documentation.
2. Audit every method against the actual execution path and tests.
3. Create docs/methodology-implementation-audit.md.
4. Present the audit and planned changes before modifying production code.
5. Reuse existing objects and implement only confirmed gaps.
6. Run project-native tests and report reused, changed, deferred and blocked components.

Repository architecture observed

Layer	Current implementation	Primary responsibility
Data pipeline	apps/data-pipeline (Python)	Download, transform, import, geography synchronization, offline model refresh
Source database	PostgreSQL/PostGIS raw tables	Immutable imported transactions and property components
Analytical database	analytics schema and migrations	Contract-grain facts, outliers, anomalies, predictions and indices
API	apps/api (Go)	Validated statistics, spatial queries, review workflow and model outputs
Dashboard	apps/web (Next.js)	Maps, market summaries, filters and visual analysis

Document map

1. Repository evidence and implementation status
2. High-priority correctness gaps
3. Required Codex audit workflow
4. Validation and valuation work package
5. Building-plot and buildable-land work package
6. Robust statistics, missingness and anomaly work packages
7. Price-index and repeat-sales work package
8. Rental yield, deferred methods and definition of done
9. Ready-to-paste Codex startup instruction
10. Methodological references

1. Repository evidence and current implementation

The guide was revised after direct inspection of the private repository. The files below are the main methodology-bearing components and must be treated as existing assets, not hypothetical future modules.

Repository file	Observed role
README.md	Monorepo architecture, refresh commands and implemented analytical capabilities
database/documentation/data-dictionary.md	Raw table grain, field semantics, relationships and coordinate system
database/migrations/010_sale_package_classification.sql	Stable building-part grouping and contract-package classification
database/migrations/011_sale_facts.sql	One analytical sale row per transaction with price derivation and geography
database/migrations/012_sale_outliers.sql	Comparable-cohort IQR/MAD evaluation and representative eligibility
database/migrations/014_rental_facts.sql	One normalized rental row per agreement and rent derivation
database/migrations/015_rental_outliers.sql	Rental comparable-cohort outliers
database/migrations/016_anomaly_signals.sql	Layered anomaly score and persistent human review state
database/migrations/018_valuation_models.sql	Versioned model runs and transaction predictions
database/migrations/019_market_indices.sql	Matched repeat transactions and market-index storage
apps/data-pipeline/real_estate_etl/modeling.py	Ridge hedonic models, cross-fitted residuals, Isolation Forest and hedonic index refresh
apps/data-pipeline/tests/test_modeling.py	Direct unit coverage for percentile scoring; broader coverage must be audited
apps/api/docs/backend-api-plan.md	Statistical safeguards and implementation roadmap
apps/api/docs/anomaly-methodology.md	Anomaly signal definitions, weights, validation and review policy
apps/api/README.md	Implemented API routes and operational behavior

1.1 Existing contract-grain protection

The earlier generic guide required transaction-grain protection. The repository now has it. analytics.sale_facts and analytics.rental_facts aggregate component rows before creating one analytical row per agreement. Price/rent derivation is recorded explicitly, and unavailable amounts remain unavailable.

> **Codex must extend these facts or add purpose-specific facts. It must not introduce a raw join that copies one contract price onto every building part or parcel.**

1.2 Existing model and anomaly pipeline

- **Hedonic baseline:** log price/rent per square metre is modelled with Ridge regression by property group.
- **Cross-fitted residuals:** random five-fold predictions generate residual anomaly ranks.
- **Isolation Forest:** a secondary multivariate anomaly score is persisted and weighted below hard rules and robust cohorts.
- **Prediction persistence:** model runs, predicted values, empirical ranges and residuals are stored in analytics tables.
- **Human review:** anomaly review status and notes survive analytical refreshes.
- **Indices:** monthly constant-quality hedonic indices and annual matched-pair indices are exposed by the API.

2. Method status matrix for Codex to verify

The statuses below are high-confidence findings from inspected files. Codex must still confirm them against all current code and tests before editing.

Method	Status	Repository evidence	Codex action
Contract/component grain	COMPLETE	sale_facts and rental_facts aggregate to one row per transaction	Reuse; add regression tests when extending
Price/rent derivation metadata	COMPLETE	component, homogeneous/single-primary or unavailable states	Preserve in all new facts and APIs
PostGIS geography assignment	COMPLETE	country/region/municipality/local spatial joins	Reuse for spatial blocking and land facts
Robust IQR/MAD outliers	COMPLETE/PARTIAL	building, municipality and region cohorts with fallback	Audit thresholds, land coverage and tests
Layered anomaly queue	COMPLETE/PARTIAL	hard, robust, duplicate, spatial, residual and Isolation Forest scores	Add near duplicates and calibrated thresholds
Hedonic Ridge model	PARTIAL	implemented and persisted by property group	Correct validation, baselines, features and uncertainty
Isolation Forest	PARTIAL	implemented as a secondary signal	Prevent temporal leakage; retain secondary role
Prediction intervals	PARTIAL / CORRECTNESS RISK	empirical width exists, but final persisted range is centred on a model fitted on all rows	Use honest held-out or conformal calibration
Hedonic index	PARTIAL	constant-quality monthly counterfactual index exists	Add period coverage, vintages, uncertainty and validation
Repeat-sales method	PARTIAL / NAME MISMATCH	median annualized matched-pair return, not weighted repeat-sales regression	Rename accurately or implement recognized regression
API distributions and robust summaries	IMPLEMENTED	percentiles, histograms, medians, cleaned/raw averages	Audit publication thresholds and metadata
Bootstrap confidence intervals	MISSING	not observed in inspected analytical path	Add where stable and computationally justified
Temporal holdout	MISSING	model uses shuffled KFold	Implement forward/out-of-time evaluation
Spatially blocked validation	MISSING	geography exists but is not used as model holdout	Implement and persist split metadata
Nonlinear ML benchmark	MISSING	only Ridge and Isolation Forest observed	Benchmark after validation framework
Quantile regression	MISSING	roadmap only	Optional after baseline; useful for premium/low-price segments
Missingness analysis	MISSING	imputation exists, but no coverage/sensitivity report observed	Add profiles, indicators and sensitivity checks
Buildable-land analysis	MISSING	land is one broad property group; readiness codes are not a first-class fact	Implement codes 1, 2 and 3 at contract grain
Near-duplicate detection	MISSING	exact analytical duplicate candidate partitions exist	Add explainable near-duplicate candidates
Moran's I / residual spatial diagnostics	MISSING	roadmap only	Add after correct spatial validation

Method	Status	Repository evidence	Codex action
Rental yield	MISSING	roadmap mentions indicative yield; no implemented route observed	Add segment-level gross estimate only
Hierarchical models	DEFERRED	explicitly a future refinement	Implement only with clear uncertainty and benefit
Bubble/cycle tests	DEFERRED_BY_DATA	not appropriate for short or unstable series	Require long quarterly history and fundamentals

3. High-priority correctness gaps

3.1 Random cross-validation is not sufficient for market deployment

modeling.py uses shuffled KFold. This is acceptable for an early cross-fitted anomaly experiment, but it is not an honest estimate of future or new-area valuation performance. Nearby and future transactions can appear in training and validation simultaneously.

- Add a latest-complete-period holdout for headline metrics.
- Add forward-chaining evaluation where history permits.
- Add spatial blocking by municipality, statistical region or a documented coarse spatial cell.
- Persist train/validation dates, spatial group and split version with every model run.
- Continue cross-fitting anomaly residuals, but do not let future data score historical observations when historical reproducibility is claimed.

3.2 Persisted ranges require honest calibration

The current code calculates residual widths from cross-fitted predictions, but centres persisted lower and upper ranges on final-model fitted predictions over the complete training set. Coverage is also calculated against those fitted predictions. This can make the reported range look more reliable than it will be on unseen transactions.

> **Required correction: evaluate and calibrate ranges entirely from out-of-fold, out-of-time or conformal residuals. Store calibration population, coverage target and achieved held-out coverage.**

3.3 The current repeat-sales output is not the Case-Shiller method

The repository matches the same cadastral building part, applies holding-period and price-ratio filters, calculates annualized log returns, takes an annual median and accumulates it. This is a valid descriptive matched-pair index, but it is not a weighted repeat-sales regression.

- Option A: rename the method and API metadata to matched_pair_median_return.
- Option B: retain the descriptive method and add a recognized repeat-sales regression as a separate method.
- Do not silently replace the existing series without migration, API and frontend compatibility analysis.
- Expose pair count, holding-period distribution, identity completeness and sensitivity filters.

3.4 Market eligibility semantics need an explicit audit

Raw transactions contain marketability, deed and transaction-type codes. The inspected sale fact does not use marketability in its eligibility rules, while rental facts use rental_transaction_type. The earlier PDF assumed verified/provisional states; that assumption must not be hard-coded without confirming the imported codebook and source semantics.

> **Codex must derive eligibility rules from codebook_values and document every included/excluded source code. Unknown codes must remain visible and must not be guessed.**

4. Required Codex audit workflow

Phase 1 - repository understanding (no production edits)

1. Read repository instructions, documentation, build files, migrations, model refresh code, API contract and tests.
2. Trace each statistic from raw source tables through analytical facts, outlier evaluation, API query and frontend display.
3. Trace each model from SQL extraction through preprocessing, splitting, fitting, persistence, API response and reliability label.
4. Identify all migrations newer than 019 before proposing a new migration number.
5. Identify all existing tests before declaring any method untested or missing.

Phase 2 - audit document

Create docs/methodology-implementation-audit.md with these columns:

```
Method | Required now? | Status | Existing code | Existing tests | Methodological gap | Planned action
```

A method is COMPLETE only when it is connected to the production path, preserves the correct analytical grain, has tests, exposes sample/method metadata and meets the validation requirements in this guide.

Phase 3 - plan before code

- List components that are already correct and will remain unchanged.
- List components that need only tests, metadata or wiring.
- List correctness fixes separately from feature additions.
- List deferred methods and exact data requirements.
- Name expected files and database objects before implementation.

Phase 4 - implementation discipline

- Extend existing migrations/functions/modules rather than creating parallel analytical systems.
- Keep imported source tables immutable.
- Make every exclusion and model decision reproducible and versioned.
- Add focused tests with each change, then run the full relevant suite.
- Do not change public API semantics without updating OpenAPI, handlers, tests and frontend consumers.

Project-native verification commands

```
make pipeline-lint
make pipeline-test
make api-vet
make api-test
make web-lint
make web-test
make analytics-refresh
make models-refresh WORKERS=6
```

5. Work package 1 - validation, baselines and valuation ranges

5.1 Split design

Purpose	Required split	Reason
Published future-performance metric	Latest complete quarter/year held out	Prevents future observations from leaking into training
Stability over time	Forward/rolling evaluation	Shows whether error changes across market regimes
Generalisation to new areas	Blocked municipality/region/spatial-cell holdout	Prevents nearby properties from making validation artificially easy
Anomaly residual score	Cross-fitting with temporal safeguards	A record must not be scored by a model trained on itself or later records

5.2 Required model comparison

1. Local median price/rent per square metre baseline, using only training observations.
2. Existing Ridge hedonic model as the transparent statistical benchmark.
3. One nonlinear benchmark only after the split framework is correct.
4. Optional CatBoost when it materially improves out-of-time and out-of-area metrics and the dependency is accepted.

Do not compare models on random splits only. Do not report one national metric without subgroup diagnostics.

5.3 Required metrics and metadata

- MAE in EUR/m2 or EUR/month as appropriate.
- Median absolute percentage error only where the denominator is valid.
- RMSE on the log target and log R-squared.
- Empirical prediction-interval coverage and average interval width.
- Metrics by property group, geography, period, area band and price band.
- Training date range, validation date range, split method, feature schema, random seed and model version.

5.4 Prediction ranges

A practical first correction is split-conformal or cross-conformal calibration on log residuals. Fit the final model on the training population, but obtain the interval width from a disjoint calibration population. Evaluate final coverage on a disjoint test population.

```
log_lower = predicted_log - calibrated_quantile
log_upper = predicted_log + calibrated_quantile
lower = exp(log_lower)
upper = exp(log_upper)
```

Calibration must be segment-aware when sample size is sufficient and must fall back in a documented hierarchy when it is not.

6. Work package 2 - building plots and buildable land

Mandatory land rule: retain every land sale in the database. Default user-facing building-plot statistics, maps, filters and lists must show only verified buildable land (currently ETN codes 1-3). Non-buildable, mixed and unknown land remains stored and auditable, but must not be mixed into default building-plot counts, prices or trends.

This is the highest-priority new market segment because it directly supports the project goal and is not represented precisely by the current broad property_group=land fact.

6.1 English terminology

- **Building plot:** a single plot intended or permitted for construction; natural European/British term.
- **Building lot:** common US term.
- **Buildable land:** the broad analytical category for land where construction is possible or expected.
- **Development land:** broader term that may include future or larger development opportunities.

6.2 ETN readiness categories

ETN code	Recommended English label	Analytical interpretation
1	Building permit available	Construction is possible and a permit exists
2	Serviced buildable land	Construction is possible and infrastructure/access is available
3	Partly serviced or future buildable land	Construction is or will be possible, but readiness is lower
1-3	Broad buildable land	All included parcels fall in codes 1, 2 or 3
1-2	Strict buildable land	All included parcels fall in codes 1 or 2

6.3 Recommended contract-grain analytical object

Before adding a migration, Codex must verify that no newer migration already implements an equivalent object. The preferred design is analytics.land_transaction_facts or an equivalent name consistent with the current project.

Field	Requirement
transaction_id	Primary key; one row per sale agreement
contract_date / dataset_year	Use contract date for market flow; retain source dataset year
parcel_count	Number of parcel rows included in the contract
land_type_codes	Distinct ETN land codes included in the package
buildability_class	Code 1, code 2, code 3, broad, strict, mixed or non-buildable
total_area	Sum compatible sold parcel area once per contract
analytical_price	Complete parcel-price sum or defensible contract total
price_per_sqm	Only when both analytical price and total area are positive
price_derivation	Component prices, land-only transaction total or unavailable
geography/location	Reuse PostGIS assignment and representative centroid
quality state	Eligibility flag and multiple explicit reasons

6.4 Eligibility and reporting rules

- Default building-plot statistics, maps, filters and lists to verified buildable classes only; non-buildable, mixed and unknown land requires an explicit audit/raw or opt-in filter.
- Keep all land transactions in the database, regardless of buildability classification.
- Use land-only contracts for clean building-plot statistics unless an explicit bundle model handles buildings.
- Aggregate compatible parcels before calculating price per square metre.
- Keep code 1, code 2 and code 3 results separately visible.
- Report unique contract count separately from parcel count.
- Do not treat a mixed agricultural/buildable/forest package as a clean building plot.
- Audit sold shares and ownership fractions before using total parcel area.
- Expose a broad codes 1-3 filter and a stricter codes 1-2 filter.
- Do not silently redefine the existing property_group=land API value; add a land-readiness dimension or documented subcategory.

6.5 Required tests

- One parcel, code 1, complete parcel price.
- Multiple parcels, all code 2, contract-level price only.
- Multiple parcels containing codes 1 and 3.
- Mixed buildable and agricultural land.
- Building plus land bundle.
- Partial ownership share.
- Missing/zero area or price.
- Contract count remains one while parcel count is greater than one.

7. Work package 3 - robust statistics and missing data

7.1 Publication policy

The API already returns robust medians, percentiles, histograms and explicit sample sizes. Codex should not replace this. It should add a consistent publication policy and uncertainty metadata.

- Define configurable minimum sample sizes for calculation and publication.
- Mark unstable cells instead of presenting false precision.
- Return population definition, eligible count, included representative count, excluded count and date basis.
- Add deterministic bootstrap confidence intervals for selected medians/changes or precompute them during refresh.
- Keep raw, representative and modelled values visibly distinct.

7.2 Missingness analysis

- Profile missingness by field, property group, geography, contract period, dataset year and source file.
- Add missingness indicators when numeric predictors are imputed.
- Never impute price, rent or target-derived values for training.
- Compare complete-case and imputed results for material predictors.
- Alert when source refreshes cause abrupt coverage changes.

7.3 Feature improvements to audit

The raw schema contains characteristics not present in the current model feature list, including floor/position, room count, new construction, furnishing, operating-cost inclusion, rental start date and potentially sold-share information. Codex should first measure coverage and leakage risk, then add only stable, interpretable features.

> **More features are not automatically better. Every feature must have a documented source, missingness rate, transformation and availability at prediction time.**

8. Work package 4 - anomaly-system improvements

The current anomaly architecture is sound: immutable source rows, named flags, weighted component scores and persistent human review. Improvements should extend that architecture.

- **Near duplicates:** add date/amount/area/location/identity tolerances and explain why a pair was suggested.
- **Temporal model integrity:** do not let future transactions influence historical residual ranks.
- **Threshold calibration:** use reviewed labels when enough exist; do not force a fixed anomaly percentage.
- **Spatial diagnostics:** distinguish missing coordinates, outside-country coordinates, boundary tolerance and claimed-location mismatch.
- **Review preservation:** refresh model signals without overwriting reviewer decisions.
- **Safety:** never describe a statistical anomaly as fraud or wrongdoing based only on ETN data.

9. Work package 5 - price-index methodology

9.1 Three-index comparison

For high-volume segments, publish or internally compare three conceptually different series:

1. Stratified median price-per-square-metre index - transparent but composition-sensitive.
2. Hedonic constant-quality index - adjusts for observed characteristics.
3. Repeat-transaction index - uses repeated observations of the same unit when data quality is sufficient.

Agreement between methods increases confidence. Divergence is a diagnostic for changing composition, sparse samples, unstable identity matching or model misspecification.

9.2 Hedonic index corrections

- Record actual observed sample size in every period, not only the reference-sample size used for prediction.
- Use a documented, stable base period and suppress periods with inadequate support.
- Store index-run/vintage metadata and preserve previous published values for revision analysis.
- Report uncertainty or sensitivity where feasible.
- Validate national and regional indices out of time before exposing granular monthly indices.
- Do not let later data silently rewrite a historical index without a visible new vintage.

9.3 Repeat-transaction decision

Option	Implementation	API/method name
Retain current method	Median annualized log return from filtered matched pairs	matched_pair_median or equivalent accurate name
Add recognized method	Bailey-Muth-Nourse / Case-Shiller weighted repeat-sales regression	repeat_sales_regression
Comparison mode	Store both methods with pair counts and diagnostics	Separate method values; never merge silently

The recognized regression should include pair construction, arm-length/eligibility rules, minimum holding period, duplicate/correction handling, heteroskedasticity treatment and minimum pair counts. It should not be implemented merely to satisfy a label if identity quality is insufficient.

9.4 Index data-vintage schema

Codex should consider adding an index run identifier or append-only vintage table instead of deleting every prior index row. A minimal vintage records source snapshot, refresh time, methodology version, base period, model run and period-level sample counts.

10. Work package 6 - nonlinear ML benchmark

A nonlinear benchmark is useful, but it is not the first missing piece. Correct splits and honest uncertainty come first.

- Keep Ridge as the transparent benchmark.
- Train a nonlinear candidate under the same temporal and spatial splits.
- Use identical eligible populations and metrics for comparison.
- Adopt the nonlinear model only when improvement is material and stable by subgroup.
- Persist feature schema, hyperparameters, dependency version and random seed.
- Expose explanations and comparable observations, not only a point estimate.
- Do not use the anomaly Isolation Forest as a valuation model.

CatBoost is a strong candidate because it handles categorical variables and missing values well. It should remain an optional ML dependency until measured benefit justifies adoption.

11. Work package 7 - modelled gross rental yield

Sale and rental rows generally do not represent the same property and period. Yield must therefore be a segment estimate, not a claimed property return.

```
estimated_gross_yield = 12 * representative_monthly_rent / representative_sale_price
```

- Match sale and rental cohorts by geography, property group, size, age and period.
- Return sale and rental sample sizes separately.
- Use robust representative values and uncertainty ranges.
- Label the result modelled gross market-segment yield.
- Do not call it net yield without vacancy, maintenance, operating expenses, taxes and financing.

12. Methods to defer

Method	Required evidence before production use
Hierarchical / partial pooling	Clear small-area benefit, exposed assumptions and calibrated uncertainty
GWR / MGWR	Adequate local density, stable bandwidth selection and spatial validation
ML chained price index	At least eight complete stable quarters and reproducible adjacent-period evaluation
Change-point detection	Sufficient long, complete series and protection against one-month noise
Bubble / explosive-root tests	Long quarterly price/fundamental series and a documented research design
Causal monetary-policy analysis	Identified shocks, macro data and a causal econometric design

13. Definition of done

- The audit document maps every method to actual repository files and tests.
- No equivalent implementation has been duplicated.
- Temporal and spatial leakage tests pass.
- Prediction ranges achieve reported held-out coverage.
- Building-plot analysis distinguishes codes 1, 2 and 3 and remains one row per contract.
- Contract count and parcel count are separately reported.
- Index methods are named accurately and preserve vintages/revisions.
- Every published result includes population, sample, date, geography, derivation and version metadata.
- Raw ETN tables remain immutable.
- OpenAPI, Go handlers/repositories, SQL, Python models and frontend behavior remain aligned.
- All relevant project-native tests pass.
- The final Codex report identifies reused, changed, deliberately unchanged, deferred and blocked components.
- Every land sale remains stored, while default user-facing building-plot results contain only verified buildable classes.

14. Ready-to-paste Codex startup instruction

```
Read `docs/ETN_Real_Estate_Analyser_Repository_Specific_Guide_v2.pdf`
and execute `docs/CODEX_REPOSITORY_AUDIT_AND_IMPLEMENTATION_v2.md`.

Work audit-first. Do not edit production code until you have created
`docs/methodology-implementation-audit.md` and presented the implementation plan.

Treat the repository as an existing analytics platform, not a greenfield project.
Reuse the current analytical facts, robust outliers, anomaly queue, model tables,
API patterns and tests. Implement only confirmed methodological gaps.

Prioritize:
1. temporal and spatial validation;
2. honest prediction-range calibration;
3. buildable-land analysis for ETN codes 1, 2 and 3, while retaining every land sale and showing only
verified buildable land by default;
4. accurate repeat-sales method naming or regression implementation;
5. index vintages, uncertainty and sample metadata;
6. missingness, near-duplicate and publication-policy improvements.

Run all relevant Python, database, Go and frontend tests and report every reused,
changed, deferred and blocked component.
```

15. Repository-specific cautions

- The current repository is small enough to audit completely. Do not rely only on file-name search or documentation claims.
- Check for migrations newer than 019 before selecting a migration number.
- A similarly named function is not proof that a method is complete; trace its production use and tests.
- Do not run destructive refreshes against production data.
- Do not infer legal status, fraud, official valuation or construction permission beyond what the source codebook supports.

16. Methodological references

[1] Eurostat, ILO, IMF, OECD, UNECE and World Bank (2013). Handbook on Residential Property Price Indices.

[2] Rosen, S. (1974). Hedonic Prices and Implicit Markets: Product Differentiation in Pure Competition.

[3] Hill, R. J. (2011). Hedonic Price Indexes for Housing. OECD Statistics Working Papers.

[4] Herath, S. and Maier, G. (2010). The Hedonic Price Method in Real Estate and Housing Market Research: A Review of the Literature.

[5] Case, K. E. and Shiller, R. J. (1987). Prices of Single-Family Homes Since 1970: New Indexes for Four Cities.

[6] S&P Dow Jones Indices (2025). S&P Cotality Case-Shiller Home Price Indices Methodology.

[7] Helbich, M., Brunauer, W., Vaz, E. and Nijkamp, P. Spatial Heterogeneity in Hedonic House Price Models: The Case of Austria.

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[9] Steurer, M. and Spiegel, S. (2026). Addressing Missing Data in Real Estate Price Indices.

[10] Liu, F. T., Ting, K. M. and Zhou, Z.-H. (2008). Isolation Forest.

Repository provenance

This revision was prepared from direct inspection of the private GitHub repository matkopolo/NepremicninskiAgent, branch main, at the observed commit 70b5ec82e964e2cf5061aa6855fb2a6b16bfe610. The high-confidence status matrix is an implementation aid, not a substitute for Codex auditing all files at execution time. The repository may change before Codex usage resets.

The original ETN row counts and raw-data field mappings remain documented in the earlier dataset guide. This version supersedes its greenfield implementation assumptions and corrects the earlier unverified requirement to separate "verified" and "provisional" transactions: Codex must first establish the actual codebook semantics.